



Comparative Study of Single and Double Gate All Around Cylindrical FET Structures for High-K Dielectric Materials

Rajveer Kaur¹ · Balwinder Singh¹

Received: 2 February 2020 / Revised: 9 October 2020 / Accepted: 2 November 2020
© The Korean Institute of Electrical and Electronic Material Engineers 2020

Abstract

The performance of conventional MOSFET deteriorates because of the short channel effects that appears when it is scaled into nm regime. The existing techniques like channel scaling, variations in the work function implemented on a MOSFET could no longer confront these limitations which demand for the necessity of some upgraded devices and materials that would overcome these shortcomings and offer ameliorate performance. Two experimentally based devices gate all around (GAA) FET and double gate all around (DGAA) FET are modeled and compared at 10 nm and 20 nm respectively. In this paper, the effect of scaling the gate length, oxide thickness and variations in the drain to source voltage utilizing two different gate dielectrics for a single and double GAA cylindrical FET on their respective device performance in terms of drive current (I_{ON}), leakage current (I_{OFF}), switching speed (I_{ON}/I_{OFF}) and subthreshold swing. The study reveals that with a thinner gate oxide, less gate length, less drain to source voltage and with an additional core gate utilizing high-k dielectric materials, the device achieves a better subthreshold slope, higher value of ON-state current, larger ON/OFF current ratio, lesser OFF-state current and lesser power consumption.

Keywords CMOS · MOSFET · Gate all around (GAA) FET · Double gate all around (DGAA) FET · Short channel effects · On current · Off current · Scaling

1 Introduction

Transistor comes into existence in early 1960s plays an essential role in the semiconductor industry. High speed, less area and low power consumptions in the portable devices become the primary concern for researchers in the semiconductor industry [1]. The reduction in the dimensions of a planar CMOS is a cumbersome task. The reduction in the dimensions of a planar CMOS provide better performance and excels the functionality of conventional MOSFET (CMOS) but scaling in nm regimes becomes a cumbersome task. It has an inferior impact on the gate control over the channel which is the main reason for performance degradation of the device as it suffers from short channel effects such as mobility degradation, variations in the threshold voltage (V_{th}), increase in the leakage current and difficulty in increasing the ON-current [1]. According to International

Technology Roadmap for Semiconductor (ITRS) roadmaps (2015) given by Semiconductor Research Corporation, the electronics industry will reach its fundamental limit by 2020s [2, 3]. The number of transistors in a single IC has already reached the maximum scaling limits over last four decades. Scaling of conventional MOSFET demands new technologies, new materials and innovative geometry structures that can ameliorate the gate control over the channel. Multigate geometries like dual gate, tri gate, gate all around geometries are the emerging structures. We approach cylindrical single and double gate all around structures. These structures provide better control of gate over the channel. Cylindrical GAA provides superior performance than single, double and triple gate geometries [5].

The basic difference between MOSFET and GAA FET is the respective geometry [4–7]. In planar MOSFET, the channel is horizontal, whereas the channel of GAA structure is cylindrical. In this, the gate surrounds the entire channel which improves its ability to control the flow of current; it lowers down the corner effects, reduces short channel effects (SCE).

✉ Balwinder Singh
Balwinder@cdac.in

¹ ACS Division, Centre of Development of Advanced Computing, Mohali 160071, India

The current is pinched off under the influence of applied voltage on the gate electrode. The source to drain current is controlled by applying voltage to the gate. Cylindrical GAA FET structure is a radical inversion and a potential candidate to replace conventional MOSFET as to introduce new direction for transistor scaling. DGAA FET also utilizes cylindrical channel surrounded by a core (inner) and a shell (outer) cylindrical gate. [8–12] The existence of two gates contributes an ultimate immunity to the SCEs. In this structure both the gates control tubular silicon channel that improve its ability to enhance the current.

In Fig. 1, cylindrical transistor schematics are shown for 20 nm gate length GAA FET (a) and 20 nm gate length DGAA FET (b).

Because of scaling conventional FETs into nm regime, SiO₂ layer used as a gate dielectric is continuously becoming thin and hence the problem of leakage current and power dissipation arises which degrades device performance. Moreover it will not be an easy task to scale the device after reaching a peak value, so there is a need to study some new materials that can be used in designing FET. To improve the performance of the device some high-k dielectric are explored [13, 14]. In this work HfO₂ is considered as it is a high-k dielectric material with value 25 which is 3.9 for SiO₂ [13]. Also HfO₂ forms a good electric interface with silicon and ensures lower leakage. In this paper, we study the performance of cylindrical gate all around (CGAA) and DGAA FET structures utilizing two different dielectrics to study.

different performance metrics like I_{ON} (on current), I_{OFF} (off current) and the ratio of ON current and OFF current, Subthreshold Swing (SS) and power are systematically presented. GAA and DGAA cylindrical FET are simulated by considering the effect of scaling the oxide, reducing the gate length, varying the drain to source voltage with the utilization of high-k dielectric material. Recently 22 nm technology have opted FinFETs (a Multigate technology) and is anticipated to work for channel lengths which are down to 10 nm. Beyond 10 nm regimes, GAA structures are considered as favorable multi gate technologies. So, we approached a comparative analysis for CGAA and CDGAA FET at 10 and 20 nm respectively.

Section 2 describes the device structure including the dimensions, materials used and doping concentration of CGAA (Cylindrical Gate All Around) FET and CDGAA (Cylindrical Double Gate All Around) FET. Section 3 comprises the results and the discussions and finally the concluding remarks are presented in last section.

2 Device Structure and Simulations

In Fig. 2, Cross-section view of cylindrical transistor is shown (a) GAA FET and (b) DGAA FET. The parameters used for the device simulations are listed in Tables 1 and 2 respectively. A gate length of 20 nm and 10 nm, an effective oxide thickness of 1 nm and 2 nm, supply voltage of 0.05 V and

Fig. 1 Schematics of the simulated **a** GAA FET with HfO₂ gate dielectric and **b** DGAA FET with SiO₂ gate dielectric

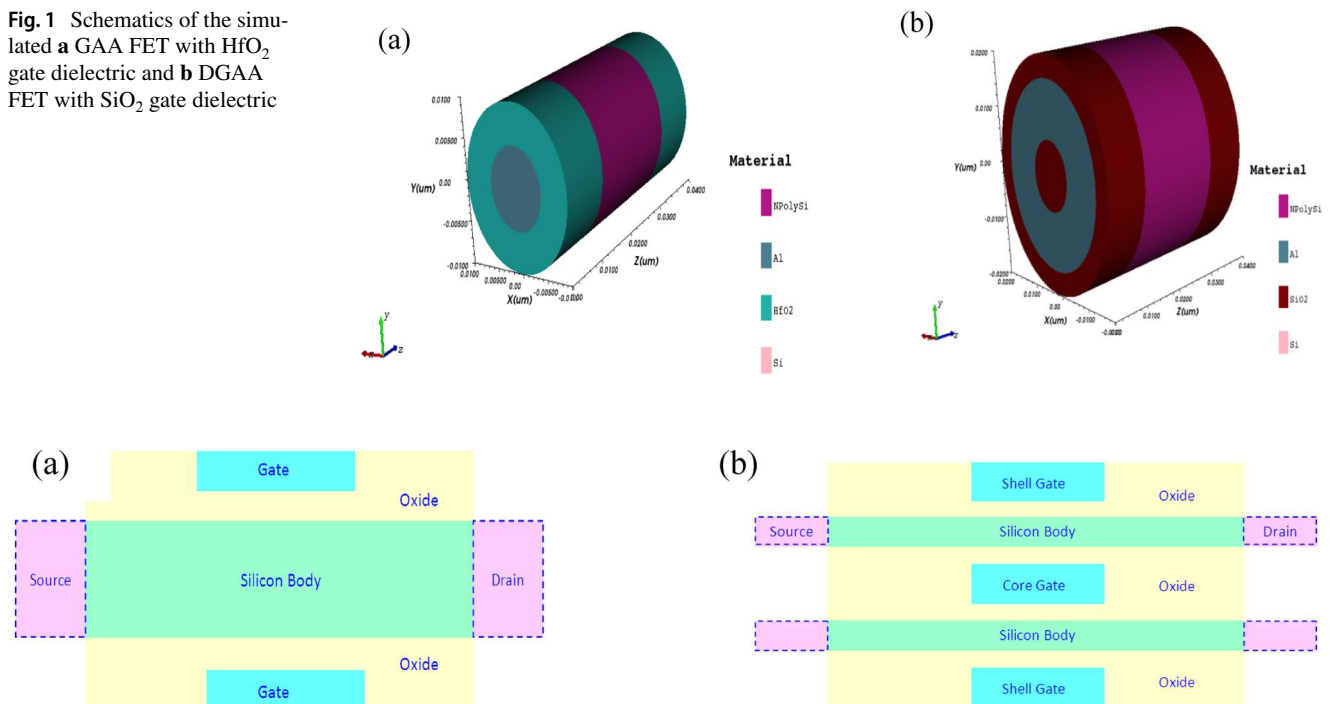


Fig. 2 Cross-section view of the simulated **a** GAA FET and **b** DGAA FET

Table 1 Parameters used for device simulation

Parameters	CGAA (cylindrical gate all around)			
	SiO ₂ (Silicon dioxide)	HfO ₂ (Hafnium oxide)		
Technology	20 nm	10 nm	20 nm	10 nm
V_{DD} , lin[V]	0.05	0.05	0.05	0.05
V_{DD} , sat[V]	0.08	0.08	0.08	0.08
L_G [nm]	20	10	20	10
Radius [nm]	10	10	10	10
Gate work function (eV)	4.45	4.45	4.45	4.45
Donor ion concentration	1E20	1E20	1E20	1E20
Acceptor ion concentration	1E18	1E18	1E18	1E18

Table 2 Parameters used for device simulation

Parameters	CDGAA (cylindrical double gate all around)			
	SiO ₂ (Silicon dioxide)	HfO ₂ (Hafnium oxide)		
Technology	20 nm	10 nm	20 nm	10 nm
V_{DD} , lin[V]	0.05	0.05	0.05	0.05
V_{DD} , sat[V]	0.08	0.08	0.08	0.08
L_G [nm]	20	10	20	10
Radius [nm]	20	20	20	20
Gate work function (eV)	4.7	4.7	4.7	4.7
Donor ion concentration	1E20	1E20	1E20	1E20
Acceptor ion concentration	1E15	1E15	1E15	1E15

0.08 V utilizing two different gate dielectrics have been used in simulations.

3 Results and Discussions

In this section, the comparative study of single and double GAA FET showing the effect of varying the gate length, scaling the gate oxide thickness (T_{OX}) and varying the drain to source voltage (V_{DS}) have been discussed for SiO₂ and HfO₂ gate dielectrics. The simulations are carried out using the device simulator visual TCAD tool, which is the latest GUI (Graphical User Interface) for Genius device simulations. The graphical data and spreadsheets of data obtained are studied considering various parameters like modifying voltages, work functions. The obtained data is analyzed using MS Excel also.

3.1 Effect of Scaling the Gate Length (L) on Drain Current

The transfer characteristics of single and double GAA cylindrical FET has been computed with variations in the

Table 3 Figure of merit for GAA and DGAA FET with SiO₂ as gate dielectric at 10 nm and 20 nm

Parameters	GAA FET (SiO ₂)		DGAA FET(SiO ₂)	
	10 nm	20 nm	10 nm	20 nm
I_{ON}	2.38E−05	1.25E−05	0.0001244	9.58E−05
I_{OFF}	9.56E−08	1.41E−09	2.03E−05	2.91E−07
I_{ON}/I_{OFF}	0.02E+03	0.88E+04	5.40E+06	3.29E+02
SS	149	100	210	189
$P_{(LEAKAGE)}$	4.78E−09	7.05E−11	1.62E−07	1.45E−08
$P_{(CONSUMED)}$	1.19E−06	6.23E−07	9.45E−06	4.79E−06

Table 4 Figure of merit for GAA and DGAA FET with HfO₂ as gate dielectric at 10 nm and 20 nm

Parameters	GAA FET (HfO ₂)		DGAA FET (HfO ₂)	
	10 nm	20 nm	10 nm	20 nm
I_{ON}	4.80E−05	2.38E−05	0.0001244	0.0001535
I_{OFF}	6.48E−09	4.90E−09	2.03E−05	2.69E−08
I_{ON}/I_{OFF}	0.74E+04	0.48E+05	5.40E+06	5.70E+08
SS	80	70	210	93
$P_{(LEAKAGE)}$	1.59E−10	2.45E−11	1.62E−07	1.34E−09
$P_{(CONSUMED)}$	2.84E−06	1.19E−06	9.45E−06	7.68E−06

gate length from 20 nm to 10 nm keeping $T_{OX} = 2$ nm and $V_{DS} = 0.05$ V. The V_{GS} is swept in interval ($0 < V_{GS} < 1$ V) with a step of 0.1 V. The simulation results for both the devices with SiO₂ and HfO₂ for cylindrical geometry have been presented in Tables 3 and 4 respectively.

Figure 3 shows the graphical variations of drain current (I_{DS}) w.r.t gate voltage (V_{GS}) for GAA and DGAA FET for SiO₂ and HfO₂ at 10 nm and 20 nm gate dielectric respectively. It can be noted from the simulation results that the current is maximum in case of DGAA FET with HfO₂ gate dielectric at 10 nm. Simulation results reveal that current is maximum in case of DGAA FET with HfO₂ gate dielectric at reduced gate length i.e. at 10 nm. When the gate length is reduced, the control of gate upon the channel also improves. The study also revealed that with utilization of high-k dielectric and an additional core gate, the drive current (I_{ON}) can be increased. HfO₂ being a high-k dielectric forms a good interface with silicon and ensures lower leakage and it also have the tendency to crystallize and have a higher electronic defect concentration. The value of Subthreshold Swing (SS) is higher in case of DGAA FET at 10 nm which reveals that with scaling of technology more gate voltage is required to change the drain current. The leakage power increases with technology scaling. The presence of extra gate provides better control over short channel effects. HfO₂ can be used in place of SiO₂ in case to reduce the leakage power.

Fig. 3 Effect of scaling the gate length on drain current for GAA and DGAA FET with **a** SiO₂ as gate oxide and **b** HfO₂ as gate oxide

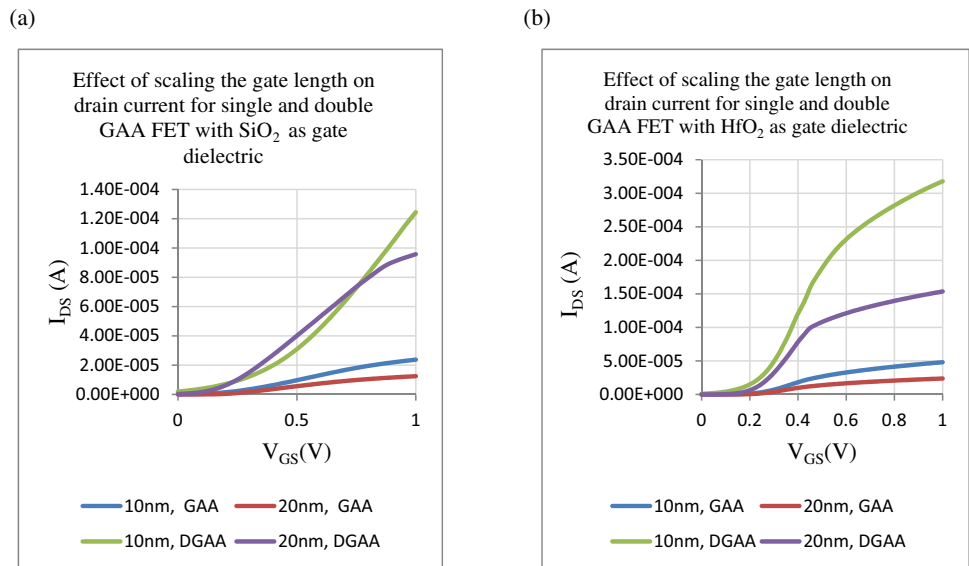


Table 5 Figure of merit for GAA and DGAA FET with SiO₂ as gate dielectric for $T_{ox} = 1$ nm and 2 nm

Parameters	GAA FET (SiO ₂)		DGAA FET(SiO ₂)	
	$T_{ox} = 1$ nm	$T_{ox} = 2$ nm	$T_{ox} = 1$ nm	$T_{ox} = 2$ nm
I_{ON}	1.56E-05	1.25E-05	0.0001756	0.000124
I_{OFF}	7.01E-10	1.41E-09	1.50E-07	2.03E-06
I_{ON}/I_{OFF}	0.22E+05	0.88E+04	1.71E+03	6.12E+01
SS	120	70	190	210
$P_{(LEAKAGE)}$	1.02E-09	4.78E-09	7.25E-09	1.62E-10
$P_{(CONSUMED)}$	1.54E-06	1.19E-06	8.78E-06	9.45E-06

Table 6 Figure of merit for GAA and DGAA FET with HfO₂ as gate dielectric for $T_{ox} = 1$ nm and 2 nm

Parameters	GAA FET (HfO ₂)		DGAA FET(HfO ₂)	
	$T_{ox} = 1$ nm	$T_{ox} = 2$ nm	$T_{ox} = 1$ nm	$T_{ox} = 2$ nm
I_{ON}	2.72E-05	2.38E-05	0.0003261	0.0003178
I_{OFF}	4.25E-10	4.90E-10	2.87E-07	9.79E-07
I_{ON}/I_{OFF}	0.64E+05	0.48E+05	1.11E+05	3.24E+02
SS	149	80	120	161
$P_{(LEAKAGE)}$	1.59E-10	3.24E-10	1.44E-10	4.89E-08
$P_{(CONSUMED)}$	2.84E-06	2.40E-06	1.63E-05	1.59E-05

3.2 Effect of Scaling the Oxide Thickness (T_{ox}) on Drain Current

Here, the transfer characteristics been computed with the variations in the oxide thickness. The oxide thickness has been scaled from 2 to 1 nm for GAA and DGAA FET cylindrical structures at 10 nm technology for $V_{DS} = 0.05$ V. The V_{GS} is swept in the interval ($0 < V_{GS} < 1$) with a step of 0.1 V. The simulation results for GAA and DGAA FET with SiO₂ and HfO₂ gate dielectrics are presented in Tables 5 and 6 respectively.

Figure 4 shows the graphical variations of I_{DS} w.r.t V_{GS} for GAA and DGAA FET for SiO₂ and HfO₂ respectively. The calculated data reveals that the drive current is maximum in case of DGAA FET with oxide thickness of 1 nm which means that with decrease in the oxide thickness, there is a rise in the drive current and a decrease in leakage power. HfO₂ in this case also proved to be an effective dielectric. HfO₂ being a high-k dielectric provides more current and have lower values for leakage current and leakage power. The minimum thickness of gate oxide for outer and inner

gate provides a better leakage current with an effective increase in the drive current. With the utilization of additional core gate, high-k dielectric and scaling down the oxide thickness, the effects of SCE's can be controlled.

3.3 Effect of Variation in the Drain to Source Voltage (V_{DS}) on Drain Current (I_{DS})

In another study of single and double GAA FET, the variations in the V_{DS} is taken into consideration, where transfer characteristics have been computed at $T_{ox} = 2$ nm, gate length of 10 nm. The V_{GS} is swept in the interval ($0 < V_{GS} < 1$) with a step of 0.1 V, and results are obtained at different V_{DS} i.e. $V_{DS} = 0.05$ V and $V_{DS} = 0.08$ V. Tables 7 and 8 show the performance analysis of GAA and DGAA FET for SiO₂ and HfO₂ respectively.

Figure 5 shows the graphical variations of I_{DS} w.r.t V_{GS} for GAA and DGAA FET for SiO₂ and HfO₂ respectively. The graphical data reveals that with increase in the drain to source voltage, the drive current also increases for both single and double GAA FET. The current is higher in case

Fig. 4 Effect of scaling the oxide thickness on drain current for GAA and DGAA FET with **a** SiO₂ as gate oxide and **b** HfO₂ as gate oxide

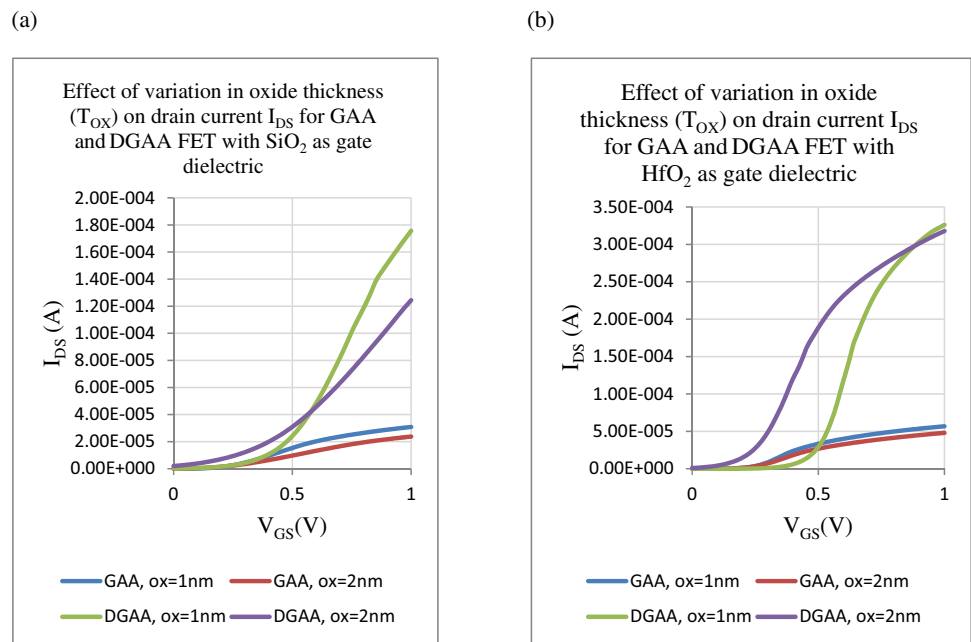


Table 7 Figure of merit for GAA and DGAA FET with SiO₂ as gate dielectric architecture at 0.05 V and 0.08 V

Parameters	GAA FET (SiO ₂)		DGAA FET (SiO ₂)	
	0.05 V	0.08 V	0.05 V	0.08 V
I_{ON}	2.38E-05	3.34E-05	0.0001244	0.0001109
I_{OFF}	9.56E-08	1.39E-07	2.03E-06	1.26E-06
I_{ON}/I_{OFF}	0.29E+03	2.40E+02	6.12E+06	0.000885E+06
SS	49	90	210	208
$P_{(LEAKAGE)}$	4.78E-09	1.11E-08	1.62E-07	6.31E-08
$P_{(CONSUMED)}$	1.19E-06	2.67E-06	9.45E-06	5.5E-06

Table 8 Figure of merit for GAA and DGAA FET with HfO₂ as gate dielectric architecture at 0.05 V and 0.08 V

Parameters	GAA FET (HfO ₂)		DGAA FET (HfO ₂)	
	0.05 V	0.08 V	0.05 V	0.08 V
I_{ON}	4.80E-05	7.28E-05	0.0003178	0.0004569
I_{OFF}	6.48E-09	8.20E-09	9.79E-07	1.44E-06
I_{ON}/I_{OFF}	0.74E+04	0.88E+04	3.24E+07	3.17E+02
SS	80	169	161	212
$P_{(LEAKAGE)}$	3.24E-10	6.56E-10	4.85E-08	1.15E-07
$P_{(CONSUMED)}$	2.40E-06	5.82E-06	1.59E-05	3.66E-05

of DGAA with HfO₂ gate dielectric at $v=0.08$ V which reveals that the drive current increases with increase in the drain to source voltage. But with increasing the V_{DS} can also impact the power consumption, leakage current and leakage current. Moreover HfO₂ proved to be more

efficient over SiO₂ as it provide more value of drive and have lesser leakages with compared to SiO₂.

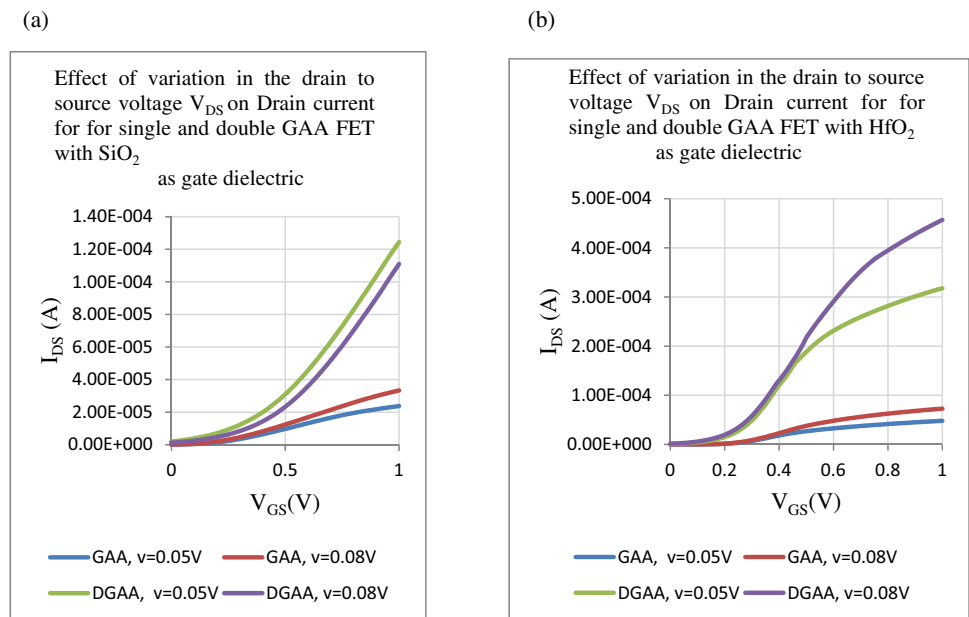
4 Conclusion

In this paper cylindrical FET with gate going around it with different configurations i.e. Gate All Around and Double Gate All Around cylindrical FET utilizing two different dielectric materials have been investigated at 10 nm and 20 nm gate length. The effect of variation in the various parameters like variations in the gate length, dielectric material, variations in the thickness of oxide, variations in the drain to source voltage are observed. The proposed structures and analysis reveals that the gate all around structures is the promising candidates to extend the downscaling of CMOS into nm regimes. The proposed structures of GAA and DGAA FET's at 10 nm have shown better electrostatic control of the gate over the channels than that of 20 nm. Moreover, the presence of extra core gate and high-k dielectric has a significant performance, leakage current, i.e. DGAA FET with HfO₂ has shown additional endurance against SCE's as compared to GAA FET.

The variations in the gate length, oxide thickness and drain to source voltage provides effective results at gate length of 10 nm, oxide thickness of 1 nm and drain to source voltage of 0.05 V.

With the scaling of gate length from 20 nm to almost half i.e. 10 nm, significant increase in I_{ON} current, I_{ON}/I_{OFF} ratio with decrease in leakage power is observed. With the downscaling of oxide thickness, effective increase I_{ON} , I_{ON}/I_{OFF} has been observed with a rise in SS also which can be

Fig. 5 Effect of variation in the drain to source voltage V_{DS} on Drain current for single and double GAA FET with **a** SiO_2 and **b** HfO_2 as gate dielectric respectively



controlled by utilizing an additional core gate with high-k dielectric. Utilization of high-K dielectric (HfO_2) in DGAA provides more endurance against SCE'S and can help- in reducing the SCE.

HfO_2 being a high-k dielectric is effective in controlling the leakage power and leakage current.

References

1. B. Buvaeswari, A survey on multi gate MOSFETS. Int. J. Innov. Res. Sci. Eng. Technol. 3, Special Issue (2014)
2. International Technology Roadmap for Semiconductors (ITRS): 201 http://www.semiconductors.org/main/2015_international_roadmap_for_semiconductors_itrs/
3. D. Jamwal, D. Dass, R. Praher, R. Vaid, *Impact of Scaling Gate Oxide Thickness on the Performance of Silicon Based Triple Gate Rectangle NWFET* (Springer, Switzerland, 2014)
4. M. Karbalaie, D. Dideban, H. Heidari, A sectorial scheme of gate-all-around field effect transistor with improved electrical characteristics. Ain Shams Eng. J. (2020)
5. P. Vimala, N.V. Rakshit, Comparative study of multigate nanowire FET transistors. J. Nanotechnol. Nano Eng. 5(3), 1–9 (2019)
6. R. Gupta, D. Dass, R. Praher, R. Vaid, *Impact of Silicon Body Thickness on the Performance of Gate-All-Around Silicon Nanowire Field Effect Transistor* (Springer, Switzerland, 2014)
7. A. Sharma, S. Akashe, Article: performance analysis of gate-all-around field effect transistor for CMOS nanoscale devices. Int. J. Comput. Appl. 84(10), 44–48 (2013)
8. R. Ambika, R. Srinivasan, Comparative performance analysis of gate inside, gate outside and gate-inside & outside cylindrical junctionless silicon nanotube FET. in *International Conference on Information, Communication and Embedded Systems (ICICES 2017)*
9. A. Agarwal, B.P. Swain, Analytical modeling of surface potential for cylindrical gate-all-around in as nanowire FET. in *IEEE International Conference on Advanced Computational and Communication Paradigms (ICACCP-2017)*
10. N.M. Mahmud Hossain, S. Quader, A.B. Siddi, M.I. Bahar Chowdhury, TCAD based performance analysis of junctionless cylindrical double gate all around FET up to 5 nm technology node. in *20th International Conference of Computer and Information Technology (ICCIT)*, 22–24 December, 2017
11. S. Quader, A.B. Siddik, N.M. Mahmud Hossain, M.I. Bahar Chowdhury, Channel engineered cylindrical double gate all around FET for low power VLSI applications. in *2018 International Conference on Computer, Communication, Chemical, Material and Electronic Engineering (IC4ME2)*
12. A. Kumar, S. Bhushan, P.K. Tiwari, A threshold voltage model of silicon-nanotube based ultra-thin double gate-all-around (DGAA) MOSFETs incorporating quantum confinement effects. IEEE Trans. Nanotechnol. 16(5), 868–875 (2016)
13. J. Robertson, High dielectric constant oxides. Eur. Phys. J. Appl. Sci. 28, 265–291 (2004)
14. A. Kumar, S. Bhushan, P.K. Tiwari, *Analytical Modeling of Sub-threshold Characteristics of Ultrathin Double Gate-All-Around (DGAA) MOSFETs Incorporating Quantum Confinement Effects* (Elsevier, Amsterdam, 2017)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.